session 8, assignment 5: time series analysis

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setting working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

loading libraries

```
library(ggplot2)
library(ggthemes)
library(stats)
library(forecast)
library(tseries)
library(formatR)
library(papaja)
library(dplyr)
library(tidyverse)
```

import data

```
data <- read.csv("tsassign.csv")
head(data)
```

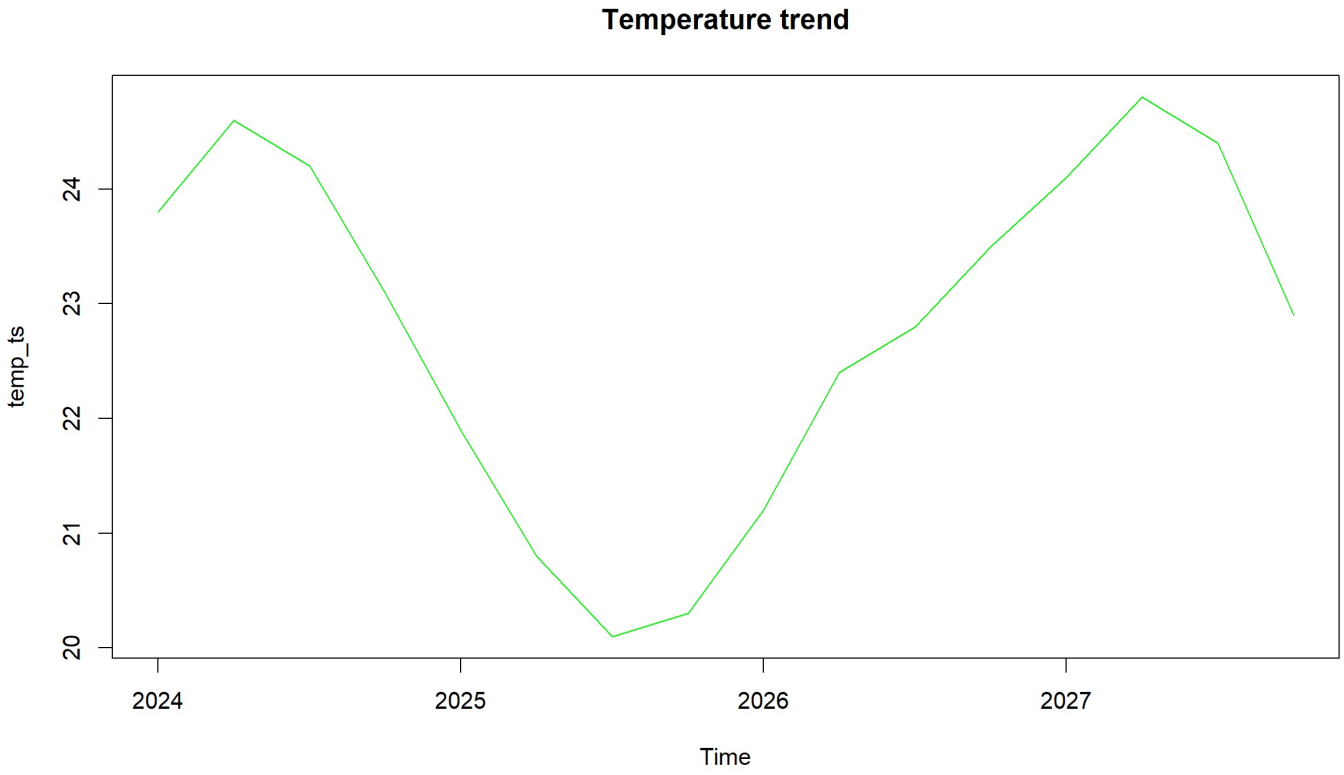
```
      Date Temperature
1 1/1/2024        23.8
2 2/1/2024        24.6
3 3/1/2024        24.2
4 4/1/2024        23.1
5 5/1/2024        21.9
6 6/1/2024        20.8
```

structure

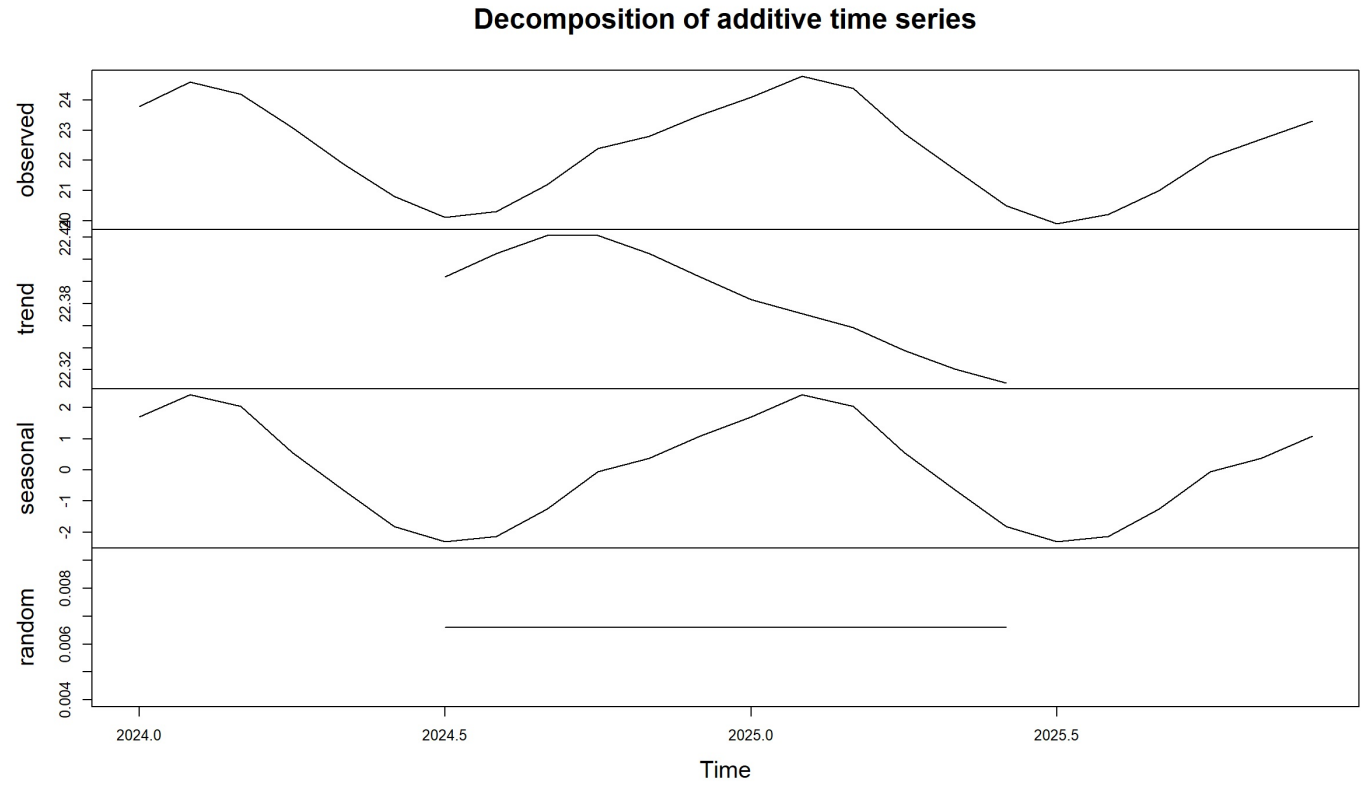
```
str(data)
```

```
'data.frame':   24 obs. of  2 variables:
 $ Date       : chr  "1/1/2024" "2/1/2024" "3/1/2024" "4/1/2024" ...
 $ Temperature: num  23.8 24.6 24.2 23.1 21.9 20.8 20.1 20.3 21.2 22.4 ...
```

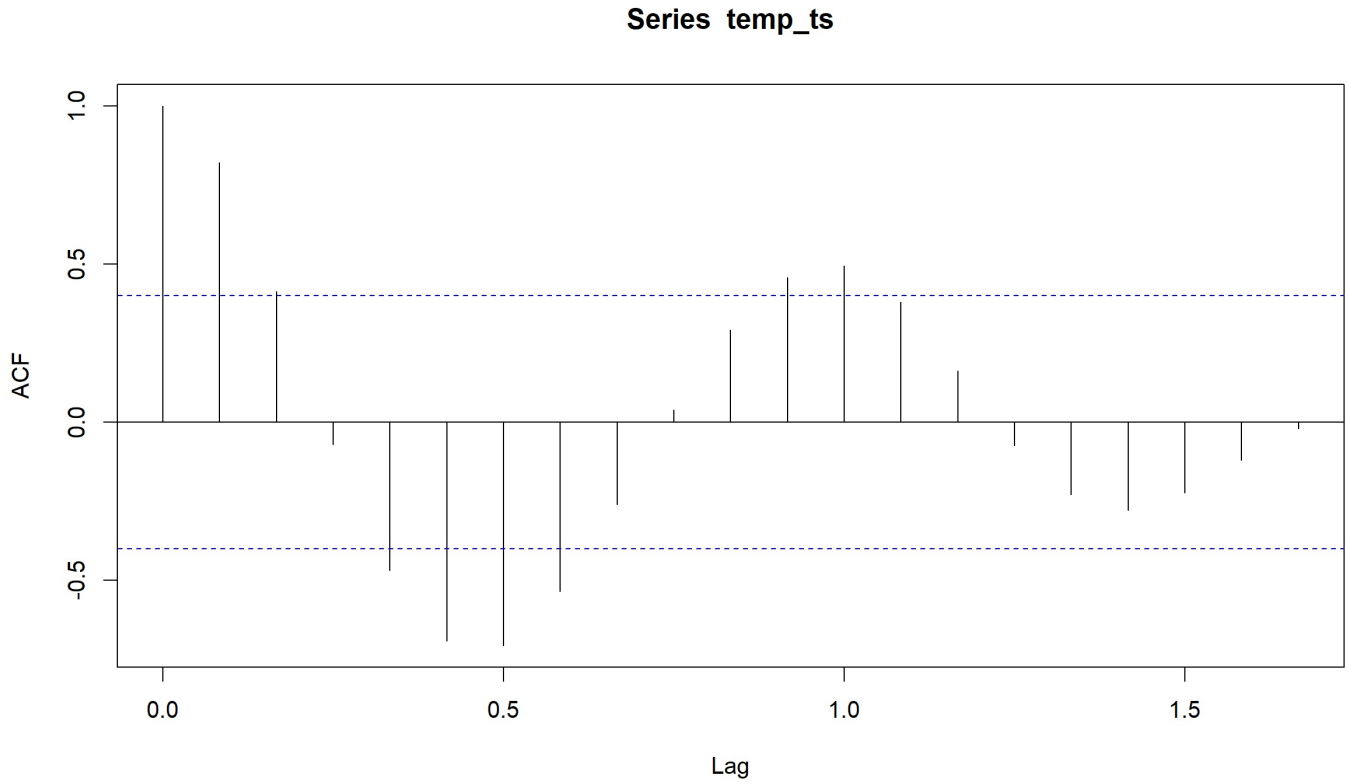
```
temp_ts <- ts(data$Temperature, start = c(2024, 1), end = c(2025, 12), frequency = 4)
plot(temp_ts, main = "Temperature trend", col = "green", )
```



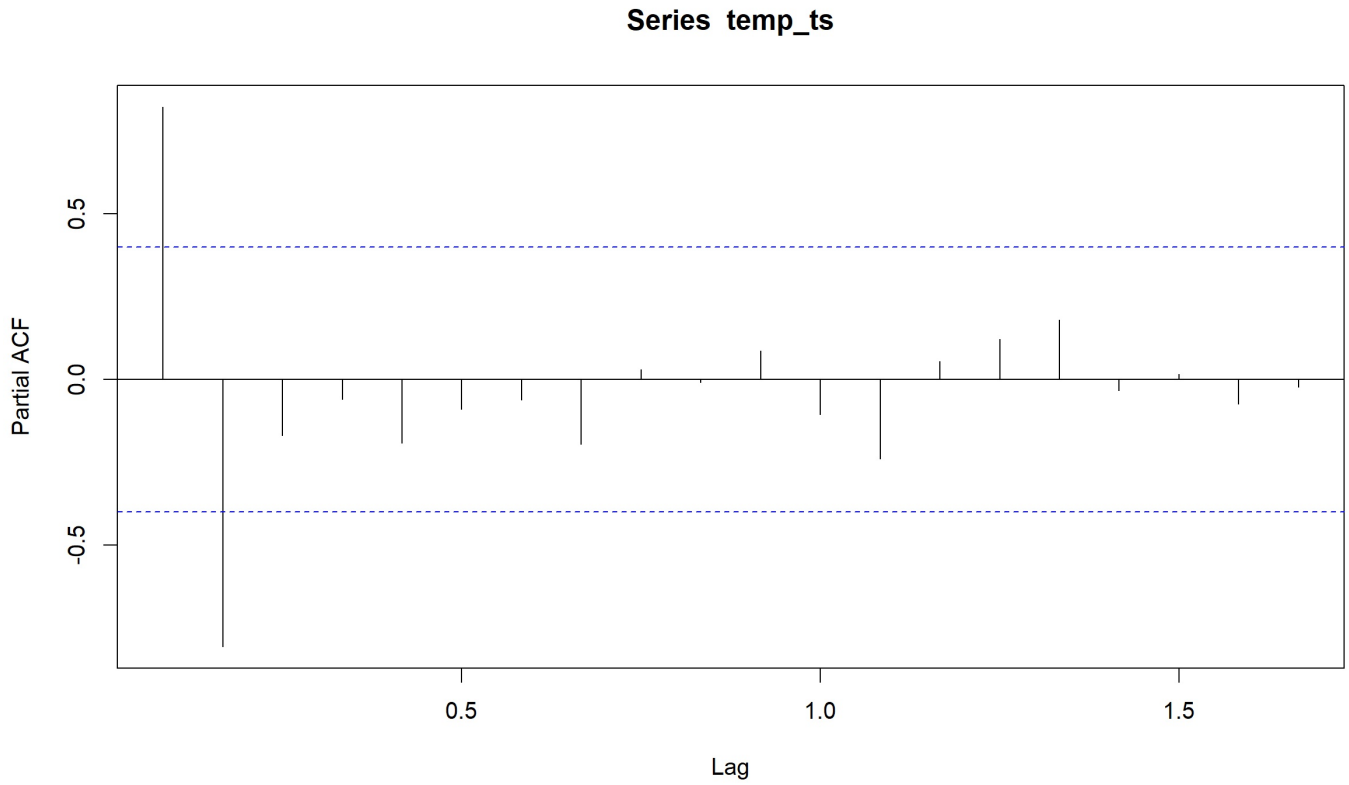

```
temp_ts <- ts(data$Temperature, frequency = 12, start = c(2024, 1))
decomposed_hgh <- decompose(temp_ts)
plot(decomposed_hgh)
```




```
acf(temp_ts, lag.max = 20)
```




```
pacf(temp_ts, lag.max = 20)
```




```
adf_reslt <- adf.test(temp_ts)
print(adf_reslt)
```

```

    Augmented Dickey-Fuller Test

data:  temp_ts
Dickey-Fuller = -3.6174, Lag order = 2, p-value = 0.04876
alternative hypothesis: stationary
```

```
arima_model1 <- auto.arima(temp_ts)
summary(arima_model1)
```

```
Series: temp_ts
ARIMA(2,0,0) with non-zero mean

Coefficients:
         ar1      ar2     mean
      1.5920  -0.9102  22.2439
s.e.  0.0688   0.0649   0.2217

sigma^2 = 0.1266:  log likelihood = -10
AIC=28.01   AICc=30.11   BIC=32.72

Training set error measures:
                     ME      RMSE       MAE        MPE     MAPE     MASE
Training set 0.01848002 0.3327948 0.2702763 0.05724893 1.178587 1.297326
                   ACF1
Training set -0.1088588
```

```
forecast_arima1 <- forecast(arima_model1, h = 10)
head(forecast_arima1)
```

```
$method
[1] "ARIMA(2,0,0) with non-zero mean"

$model
Series: temp_ts
ARIMA(2,0,0) with non-zero mean

Coefficients:
         ar1      ar2     mean
      1.5920  -0.9102  22.2439
s.e.  0.0688   0.0649   0.2217

sigma^2 = 0.1266:  log likelihood = -10
AIC=28.01   AICc=30.11   BIC=32.72

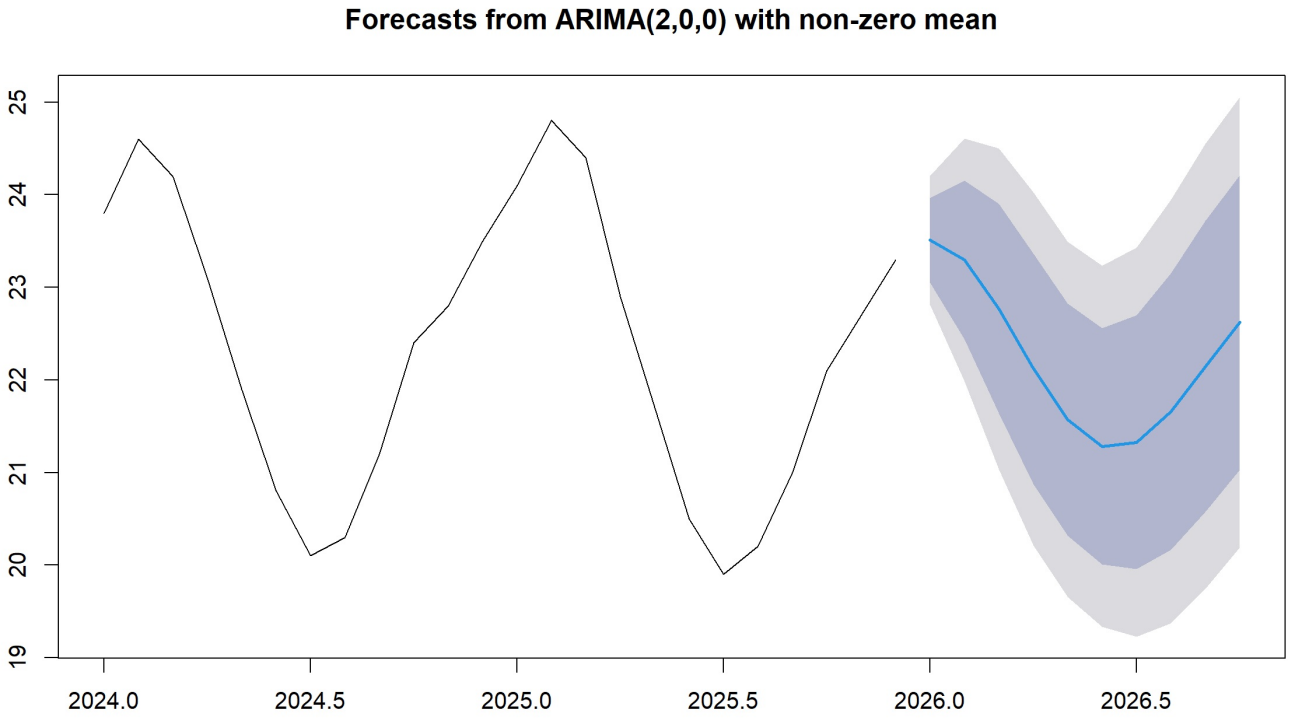
$level
[1] 80 95

$mean
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug
2026 23.51010 23.29849 22.77038 22.12223 21.57101 21.28339 21.32718 21.65868
          Sep      Oct
2026 22.14657 22.62159

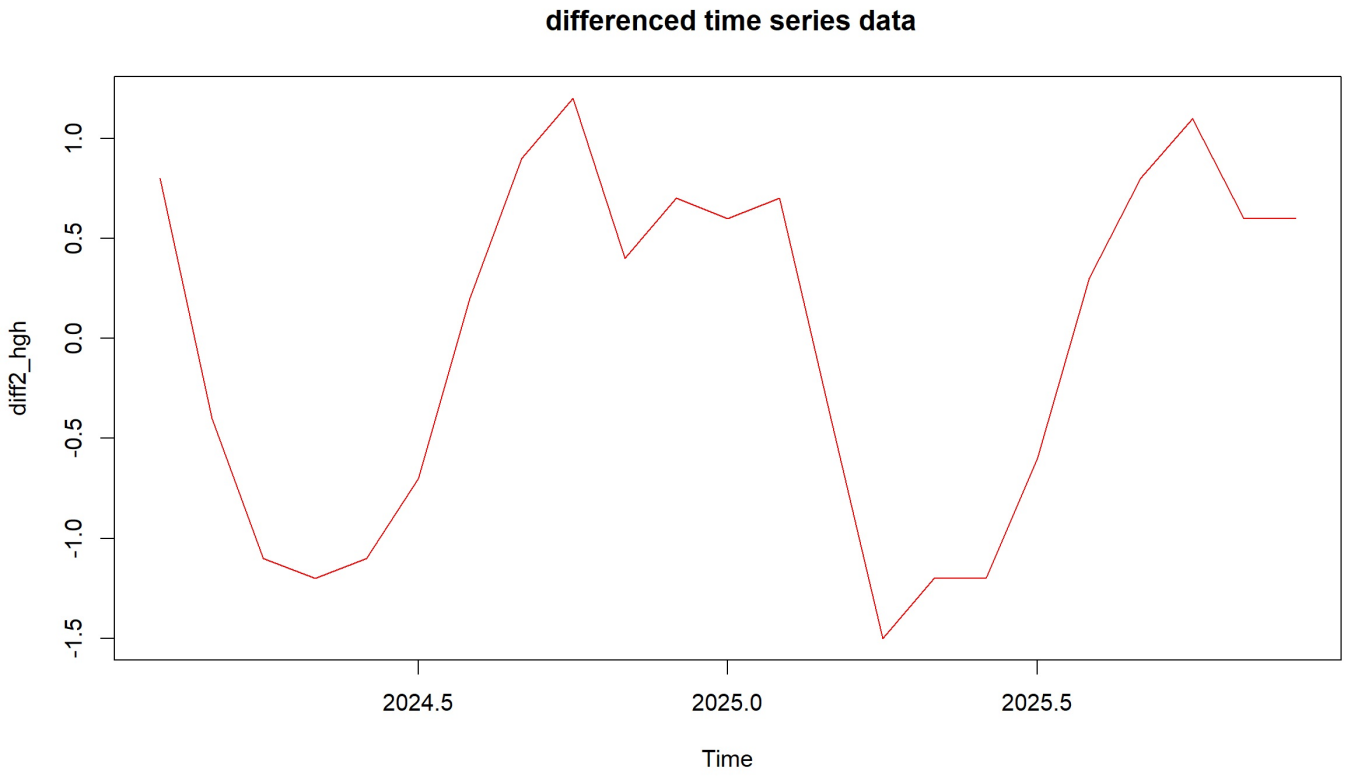
$lower
              80%      95%
Jan 2026 23.05416 22.81280
Feb 2026 22.44131 21.98755
Mar 2026 21.63757 21.03790
Apr 2026 20.87643 20.21694
May 2026 20.31607 19.65174
Jun 2026 20.00735 19.33186
Jul 2026 19.95467 19.22811
Aug 2026 20.16297 19.37119
Sep 2026 20.57379 19.74121
Oct 2026 21.03162 20.18994

$upper
              80%      95%
Jan 2026 23.96604 24.20740
Feb 2026 24.15567 24.60944
Mar 2026 23.90320 24.50287
Apr 2026 23.36803 24.02751
May 2026 22.82596 23.49029
Jun 2026 22.55943 23.23492
Jul 2026 22.69969 23.42625
Aug 2026 23.15439 23.94617
Sep 2026 23.71936 24.55194
Oct 2026 24.21156 25.05324
```

```
plot(forecast_arima1)
```




```
diff2_hgh <- diff(temp_ts)
plot(diff2_hgh, main = " differenced time series data", col = "red")
```




```
adf.test(diff2_hgh)
```

```
    Augmented Dickey-Fuller Test

data:  diff2_hgh
Dickey-Fuller = -3.0102, Lag order = 2, p-value = 0.1875
alternative hypothesis: stationary
```